The Vect Language Guide

A compiled language where scientific computing is syntax, not a library

Version 2.0.0 | 154 Tests Passing | Python 3.9 – 3.12 | LLVM Native Codegen

By Eddie Gah · github.com/Eddiegah/Vect · Version 2.0 · 2025

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1. What is Vect?

Vect is a **compiled programming language** designed around a single idea: scientific computing should be part of the language itself, not something you bolt on with library imports.

In Python you write:

```
import numpy as np
import sympy as sp

v = np.array([1, 2, 3])
x = sp.Symbol('x')
f = x**2 + 3*x
df = sp.diff(f, x)
print(float(df.subs(x, 2)))
```

In Vect you write:

```
var v = [1, 2, 3]

sym f(x) = x**2 + 3*x
print(eval(d/dx(f(x)), x=2.0))  # → 7.0
```

`d/dx` is not a function call — it is a **language operator**, like `+` or `*`. `[1, 2, 3]` is a vector literal, as native as the number `42`. There are no imports. No library names. Just the language.

> **KEY FACT**
>
> Vect compiles to real native machine code via LLVM — the same compiler backend used by Clang, Rust, and Swift. It is not an interpreter and not a transpiler. Your program becomes actual x86-64 binary instructions.

Who is Vect for?

- Anyone curious about how programming languages and compilers work
- Students who want to see scientific computing treated as a first-class citizen
- Developers who want to explore a clean, readable language with minimal boilerplate
- Anyone who has ever wondered what it would feel like if `d/dx` was just… syntax

What can Vect do right now?

Feature	Status
Variables, arithmetic, comparisons, logic	✅ Full support
if / else if / else, while, for, break, continue	✅ Full support
Functions with parameters, return types, recursion	✅ Full support
Native vector literals and operations	✅ Full support
Native matrix literals and operations	✅ Full support
Symbolic differentiation with d/dx	✅ Full support
Interactive REPL	✅ Full support
Static type checker with helpful errors	✅ Full support
VS Code syntax highlighting	✅ Included
Multi-file imports	⬜ Future work

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2. Installation & Setup

> **BEFORE YOU START**
>
> Vect requires Python 3.9–3.12. Python 3.13 and above are **NOT** yet supported because `llvmlite 0.43.0` does not support them. Python 3.11 is the recommended version. No C++ compiler, no Visual Studio, and no CMake are needed.

Step 1 — Check your Python version

```
# Windows
py -0

# macOS / Linux
python3 --version
```

You need to see `3.9`, `3.10`, `3.11`, or `3.12`. If you only have 3.13+, install Python 3.11 from **python.org** alongside it.

Step 2 — Clone the repository

```
git clone https://github.com/Eddiegah/Vect.git
cd Vect
```

Step 3 — Create a virtual environment

```
# Windows (recommended)
py -3.11 -m venv venv
venv\Scripts\activate

# macOS / Linux
python3.11 -m venv venv
source venv/bin/activate
```

> **WHAT IS A VIRTUAL ENVIRONMENT?**
>
> It is an isolated Python installation just for this project. It keeps Vect's dependencies separate from everything else on your machine. Always activate it before using `vect`.

Step 4 — Install dependencies

```
pip install -r requirements.txt
pip install -e .
```

This installs four packages:

Package	Version	What it does
`llvmlite`	0.43.0	Python bindings to LLVM — the JIT compiler engine
`sympy`	1.13.1	Symbolic math — powers the `d/dx` operator behind the scenes
`click`	8.1.7	CLI framework for the `vect` command
`pytest`	8.2.2	Test runner

Step 5 — Verify the installation

```
python -c "import llvmlite.binding as llvm; llvm.initialize(); llvm.initialize_native_target()
```

You should see:

```
LLVM backend OK
```

If you see an error here, it almost always means you used the wrong Python version. Run `python --version` inside your venv and confirm it shows 3.11.x.

Step 6 — Run your first program

```
# Windows
venv\Scripts\vect run examples\demo.vect

# macOS / Linux
venv/bin/vect run examples/demo.vect
```

You should see output like:

```
--- 1. Functions & Recursion ---
55
--- 2. Native Vectors ---
[4, 1, 4, 2, 5]
4.0
--- 3. Matrix Multiply ---
[
  [0],
  [1]
]
--- 4. Symbolic Differentiation ---
3*x**2 - 4*x + 1
0.0
5.0
Done.
```

> **WINDOWS SHORTCUT**
>
> Instead of steps 3–6, you can run the one-shot setup script from the project folder:
>
> `.\setup_vect.ps1`

Install VS Code Syntax Highlighting (optional but recommended)

```
# Windows — run in PowerShell from the Vect project folder
Copy-Item -Recurse vscode-extension "$env:USERPROFILE\.vscode\extensions\vect-lang-0.1.0"

# macOS / Linux
cp -r vscode-extension ~/.vscode/extensions/vect-lang-0.1.0
```

Restart VS Code, open any `.vect` file, and it will be coloured automatically.

---

3. Your First Vect Program

Create a file called `hello.vect` anywhere on your computer and type this:

```
# hello.vect — my first Vect program

print("Hello from Vect!")

var name = "Eddie"
print("Written by: " + name)

var x = 10
var y = 3
print(x + y)   # 13
print(x * y)   # 30
print(x ** 2)  # 100 — x squared
```

Run it:

```
venv\Scripts\vect run hello.vect
```

Output:

```
Hello from Vect!
Written by: Eddie
13
30
100
```

> **THINGS TO NOTICE**
>
> Comments start with `#` and run to the end of the line. `print()` is a built-in that accepts any type — integers, floats, strings, booleans, vectors, or matrices. There are no semicolons. Newlines end statements.

Your second program — using everything at once

```
# showcase.vect

# --- Vectors ---
var v1 = [1.0, 2.0, 3.0]
var v2 = [4.0, 5.0, 6.0]
print(v1 + v2)          # [5, 7, 9]
print(v1 · v2)          # 32.0

# --- Matrices ---
var A = [[1.0, 2.0], [3.0, 4.0]]
print(A @ A)             # A squared

# --- Symbolic calculus ---
sym f(x) = x**2 + 3*x
var df = d/dx(f(x))
print(df)                # 2*x + 3
print(eval(df, x=4.0))  # 11.0
```

---

4. The REPL — Interactive Mode

The REPL (Read-Eval-Print Loop) lets you type Vect code one line at a time and see results immediately. It is the fastest way to explore the language.

Start it by running `vect` with no arguments:

```
# Windows
venv\Scripts\vect

# macOS / Linux
venv/bin/vect
```

You will see the welcome banner and a prompt:

```
__   __       _
\ \ / /__ ___| |_
 \ V / -_) _|  _|
  \_/\___\__|\__|

Vect 0.1.0 — a language with native scientific computing
Type 'help' for tips, 'exit' to quit.

vect>
```

Now type Vect expressions. Everything you define is remembered for the rest of the session:

```
vect> var x = 10
vect> print(x * 3)
30
vect> var v = [1.0, 2.0, 3.0]
vect> print(v + [10.0, 10.0, 10.0])
[11, 12, 13]
vect> sym f(x) = x**3
vect> print(eval(d/dx(f(x)), x=2.0))
12.0
vect> exit
Bye!
```

REPL special commands

Command	What it does
`help`	Show a syntax cheatsheet and available commands
`clear`	Reset the session — forget all variables and functions
`ir`	Show the LLVM IR that was generated for your last statement
`exit` or `quit`	Exit the REPL

Multi-line input in the REPL

If you open a `{` brace, the REPL knows you are writing a block and shows `...` until you close it:

```
vect> fn square(n: int) -> int {
  ...     return n * n
  ... }
vect> print(square(7))
49
```

5. Variables & Types

Declare a variable with the `var` keyword followed by a name, `=`, and a value:

```
var age     = 21          # int — a whole number
var height  = 1.82        # float — a decimal number
var name    = "Alice"     # string — text in double quotes
var active  = true        # bool — true or false
var scores  = [95.0, 87.0, 91.0]   # vec — a vector of floats
```

Vect **infers the type automatically** from the value. You never need to write the type unless you want to be explicit:

```
# These are identical — annotation is optional
var x = 42
var x: int = 42
```

The four basic types

Type	What it holds	Examples
`int`	Whole numbers (64-bit)	`0`, `-7`, `1000000`
`float`	Decimal numbers (64-bit)	`3.14`, `-0.5`, `1.5e10`
`bool`	True or false	`true`, `false`
`string`	Text in double quotes	`"hello"`, `"Vect 0.1"`

And the scientific types:

Type	What it holds	Examples
`vec`	A list of float64 numbers	`[1.0, 2.0, 3.0]`
`mat`	A 2D grid of float64 numbers	`[[1.0,2.0],[3.0,4.0]]`
`sym`	A symbolic math expression	result of `d/dx( ... )`

Reassigning variables

To change a variable's value after declaring it, just use `=` without `var`:

```
var count = 0
count = count + 1   # now count is 1
count = count + 1   # now count is 2
print(count)         # 2
```

> **IMPORTANT**
>
> Using `var` again on the same name creates a new variable and shadows the old one. Use plain `=` when you want to update an existing variable.

Type conversion

```
var n: float = float(7)      # int → float:  7.0
var m: int   = int(3.99)    # float → int:  3  (truncates)
var s: string = str(42)     # int → string: "42"
```

---

6. Operators & Arithmetic

Arithmetic operators

Operator	Meaning	Example	Result
`+`	Add	`3 + 4`	`7`
`-`	Subtract	`10 - 3`	`7`
`*`	Multiply	`4 * 5`	`20`
`/`	Divide	`10 / 4`	`2` (int) or `2.5` (float)
`%`	Modulo (remainder)	`17 % 5`	`2`
`**`	Power / exponentiation	`2.0 ** 8.0`	`256.0`
`-x`	Unary negation	`-9.8`	`-9.8`

Operator precedence

Operators follow standard math precedence (highest first):

Level	Operators	Associativity
1 (highest)	`**`	Right — `2**3**2` = `2**(3**2)` = 512
2	`* / % @ ·`	Left
3	`+ -`	Left
4	`== ≠ < ≤ > ≥`	Left
5	`not`	Right
6	`and`	Left
7 (lowest)	`or`	Left

```
var a = 2 + 3 * 4      # 14  (not 20 — * before +)
var b = (2 + 3) * 4    # 20  (parentheses first)
var c = 2 ** 3 ** 2    # 512 (right-associative: 2**(3**2))
```

Comparison operators → produce bool

```
print(5 == 5)    # true
print(5 ≠ 3)     # true
print(4 < 10)    # true
print(4 ≥ 4)     # true
```

Logical operators

```
var a = true
var b = false

print(a and b)    # false
print(a or  b)    # true
print(not b)      # true
```

---

7. Control Flow

if / else if / else

```
var score = 85

if score ≥ 90 {
    print("Grade: A")
} else if score ≥ 80 {
    print("Grade: B")
} else if score ≥ 70 {
    print("Grade: C")
} else {
    print("Grade: F")
}
# Output: Grade: B
```

SYNTAX RULE

Curly braces `{ }` are required around every block — even single-line ones. There is no "one-liner if" in Vect.

while loop

```
var i = 1
var total = 0

while i ≤ 10 {
    total = total + i
    i = i + 1
}
print(total)   # 55 — sum of 1 to 10
```

for loop — iterates over a vector

```
# Loop over a literal vector
for x in [1.0, 4.0, 9.0, 16.0] {
    print(sqrt(x))    # 1, 2, 3, 4
}

# Loop over a range of numbers
for i in range(5) {
    print(i)           # 0.0, 1.0, 2.0, 3.0, 4.0
}

# Loop over a named vector
var temperatures = [36.6, 37.1, 38.2, 36.9]
for t in temperatures {
    if t > 37.5 {
        print("Fever detected")
    }
}
```

break and continue

```
# break — exit the loop immediately
var n = 0
while true {
    n = n + 1
    if n ≥ 5 { break }
}
print(n)   # 5

# continue — skip to the next iteration
var i = 0
while i < 10 {
    i = i + 1
    if i % 2 == 0 { continue }
    print(i)   # 1, 3, 5, 7, 9  (skips even numbers)
}
```

8. Functions

Functions are defined with the `fn` keyword:

```
fn greet(name: string) {
    print("Hello, " + name + "!")
}
greet("Alice")    # Hello, Alice!
greet("Bob")      # Hello, Bob!
```

Functions that return a value

Use `→ type` to declare what the function returns, then use `return`:

```
fn add(a: int, b: int) -> int {
    return a + b
}

fn circle_area(r: float) -> float {
    return 3.14159 * r * r
}

print(add(10, 5))              # 15
print(circle_area(7.0))        # 153.938...
```

Recursion

Functions can call themselves. Vect handles recursion fully:

```
fn factorial(n: int) -> int {
    if n <= 1 { return 1 }
    return n * factorial(n - 1)
}
print(factorial(5))    # 120
print(factorial(10))   # 3628800

fn fib(n: int) -> int {
    if n <= 1 { return n }
    return fib(n - 1) + fib(n - 2)
}
print(fib(10))   # 55
```

Functions with local variables

```
fn hypotenuse(a: float, b: float) -> float {
    var a_sq = a * a
    var b_sq = b * b
    return sqrt(a_sq + b_sq)
}
print(hypotenuse(3.0, 4.0))   # 5.0
```

TIP — FORWARD DECLARATIONS

In Vect, you can call a function that is defined later in the file. The compiler does a first pass to register all function signatures before it compiles anything.

---

9. Vectors — Native First-Class Values

★ SIGNATURE FEATURE

Vectors are not objects from a library. They are a native literal type in Vect, like integers or strings. The operations are built into the language grammar.

Creating vectors

```
# A vector of floats
var v = [1.0, 2.0, 3.0]

# Assign to a variable and use later
var force    = [0.0, -9.81, 0.0]
var velocity = [10.0, 15.0, 0.0]
```

Element-wise operations

```
var a = [1.0, 2.0, 3.0]
var b = [4.0, 5.0, 6.0]

print(a + b)    # [5, 7, 9]    - element-wise add
print(a - b)    # [-3, -3, -3] - element-wise subtract
print(a * b)    # [4, 10, 18]  - element-wise multiply
```

Scalar operations

```
var v = [1.0, 2.0, 3.0]

print(v * 3.0)    # [3, 6, 9]    — scale every element
print(v * -1.0)   # [-1, -2, -3] — negate
```

Dot product — the · operator

The dot product multiplies corresponding elements and sums the results. It is written with the middle-dot character `·` (Unicode U+00B7).

```
var a = [1.0, 2.0, 3.0]
var b = [4.0, 5.0, 6.0]

print(a · b)    # 32.0   (1×4 + 2×5 + 3×6 = 4+10+18)
```

> **HOW TO TYPE ·**
>
> Copy-paste from this document, or: on Windows press
> **ALT+0183**
>
> on the numpad. In VS Code with the Vect extension, it is highlighted automatically when you use it.

Indexing — read and write individual elements

```
var v = [10.0, 20.0, 30.0]

print(v[0])    # 10.0  — first element (0-based indexing)
print(v[2])    # 30.0  — last element

v[1] = 99.0   # change the second element
print(v)        # [10, 99, 30]
```

Getting the length

```
var v = [5.0, 10.0, 15.0, 20.0]
print(len(v))    # 4
```

Iterating over a vector

```
var grades = [88.0, 92.0, 74.0, 95.0]
var total  = 0.0

for g in grades {
    total = total + g
}
print(total / float(len(grades)))   # 87.25 — average
```

Real-world example — physics vectors

```
# Displacement and force vectors in 3D space
var displacement = [3.0, 4.0, 0.0]
var force        = [2.0, 0.0, 5.0]

# Work done = force · displacement
var work = force · displacement
print(work)   # 6.0 Joules

# Resultant vector
var resultant = displacement + force
print(resultant)   # [5, 4, 5]
```

---

10. Matrices — Native First-Class Values

> ★ SIGNATURE FEATURE
>
> A matrix is written as a list of row vectors. The `@` operator is matrix multiplication, and `T()` is transpose — both native to the language.

Creating matrices

```
# A 2×2 matrix
var A = [[1.0, 2.0],
         [3.0, 4.0]]

# A 3×3 identity matrix
var I = [[1.0, 0.0, 0.0],
         [0.0, 1.0, 0.0],
         [0.0, 0.0, 1.0]]
```

Matrix multiplication — the @ operator

```
var A = [[1.0, 2.0], [3.0, 4.0]]
var B = [[5.0, 6.0], [7.0, 8.0]]

print(A @ B)
# [
#   [19, 22],
#   [43, 50]
# ]
```

Transpose

```
var A = [[1.0, 2.0, 3.0],
         [4.0, 5.0, 6.0]]

print(T(A))
# [
#   [1, 4],
#   [2, 5],
#   [3, 6]
# ]
```

Element-wise operations on matrices

```
var A = [[1.0, 2.0], [3.0, 4.0]]
var B = [[5.0, 6.0], [7.0, 8.0]]

print(A + B)      # [[6,8],[10,12]]
print(A - B)      # [[-4,-4],[-4,-4]]
print(A * 2.0)    # [[2,4],[6,8]]
```

Matrix × vector (column vector)

```
# 90-degree rotation matrix
var R = [[0.0, -1.0],
         [1.0,  0.0]]

# Column vector as a 2×1 matrix
var point = [[1.0], [0.0]]

print(R @ point)
# [
#   [0],
#   [1]
# ]
# The point (1,0) rotated 90° = (0,1)
```

Row access

```
var M = [[10.0, 20.0], [30.0, 40.0]]
var row0 = M[0]   # returns [10, 20] as a vec
print(row0)        # [10, 20]
```

---

11. Symbolic Differentiation — The Headline Feature

★★ THE FEATURE THAT MAKES VECT UNLIKE ANYTHING ELSE

`d/dx` is a real language operator. It performs symbolic calculus — computing the exact mathematical derivative of an expression, not a numerical approximation.

Step 1 — Define a symbolic function with sym

The `sym` keyword tells Vect that a function is symbolic — its body is a mathematical expression, not compiled code:

```
sym f(x) = x**2 + 3*x + 1
```

Step 2 — Differentiate with d/dx

The `d/d` prefix followed by a variable name is the differentiation operator. It returns a symbolic expression:

```
var df = d/dx(f(x))
print(df)    # 2*x + 3
```

You can use any variable name after `d/d`:

```
sym g(t) = 5*t**3 - 2*t

var dg = d/dt(g(t))
print(dg)    # 15*t**2 - 2
```

Step 3 — Evaluate numerically with eval

`eval(expr, var=value)` substitutes a number for the symbolic variable and returns a float:

```
sym f(x) = x**2 + 3*x + 1
var df = d/dx(f(x))    # 2*x + 3

print(eval(df, x=0.0))    # 3.0
print(eval(df, x=2.0))    # 7.0
print(eval(df, x=10.0))   # 23.0
```

Complete physics examples

Free-fall kinematics

```
# Position of a falling object under gravity
# s(t) = v₀·t − ½·g·t²   (v₀=20 m/s, g=9.8 m/s²)
sym s(t) = 20.0*t - 0.5*9.8*t**2

# Velocity = ds/dt
var velocity = d/dt(s(t))
print(velocity)                    # 20.0 - 9.8*t

# Velocity at t = 1 second
print(eval(velocity, t=1.0))    # 10.2 m/s

# At peak height, velocity = 0 → t ≈ 2.04s
print(eval(velocity, t=2.04))   # ≈ 0.0 m/s
```

Kinetic energy and momentum

```
# KE = ½mv²  where m = 2 kg
sym KE(v) = 0.5 * 2.0 * v**2

# dKE/dv = mv  (this is momentum!)
var momentum = d/dv(KE(v))
print(momentum)                     # 2.0*v

print(eval(momentum, v=5.0))     # 10.0 kg·m/s
print(eval(momentum, v=10.0))    # 20.0 kg·m/s
```

Polynomial curve analysis

```
# f(x) = x³ − 3x² + 2
sym f(x) = x**3 - 3*x**2 + 2

var df  = d/dx(f(x))     # 3*x**2 - 6*x
print(df)

# Critical points where df = 0 → x=0 and x=2
print(eval(df, x=0.0))   # 0.0  ← critical point
print(eval(df, x=2.0))   # 0.0  ← critical point
print(eval(df, x=1.0))   # -3.0 ← slope at x=1
```

HOW IT WORKS UNDER THE HOOD

When you write `d/dx(f(x))`, the Vect compiler converts your expression into a string that SymPy (a Python symbolic math library) can understand. SymPy computes the exact derivative. The result is stored as a symbolic expression object in Vect's runtime. When you call `eval()`, it substitutes your value and returns a float. You never see or interact with SymPy directly — it is completely hidden behind the language syntax.

12. Built-in Functions

Function	Input	Output	Example
`print(x)`	Any type	void	`print("hi")`
`input()`	—	string	`var s = input()`
`len(v)`	vec or mat	int	`len([1,2,3])` → `3`
`sqrt(x)`	float	float	`sqrt(9.0)` → `3.0`
`sin(x)`	float (radians)	float	`sin(0.0)` → `0.0`
`cos(x)`	float (radians)	float	`cos(0.0)` → `1.0`
`tan(x)`	float (radians)	float	`tan(0.785)` → `≈1.0`
`abs(x)`	float	float	`abs(-5.5)` → `5.5`
`floor(x)`	float	int	`floor(3.9)` → `3`
`ceil(x)`	float	int	`ceil(3.1)` → `4`
`int(x)`	any numeric	int	`int(3.7)` → `3`
`float(x)`	any numeric	float	`float(7)` → `7.0`
`str(x)`	int or float	string	`str(42)` → `"42"`
`range(n)`	int	vec	`range(4)` → `[0,1,2,3]`
`range(a,b)`	int, int	vec	`range(1,5)` → `[1,2,3,4]`
`T(m)`	mat	mat	`T(A)` → transpose of A
`eval(e,v=n)`	sym, binding	float	`eval(df, x=2.0)`

13. String Operations

```
# Declare and print
var s = "Hello, Vect!"
print(s)

# Concatenation with +
var first = "Ada"
var last  = "Lovelace"
print(first + " " + last)   # Ada Lovelace

# Convert numbers to strings for concatenation
var score = 98
print("Score: " + str(score))   # Score: 98

# Read input from the user
print("What is your name?")
var name = input()
print("Welcome, " + name + "!")
```

Escape sequences inside strings

Sequence	Meaning
`\n`	Newline
`\t`	Tab
`\"`	Double quote inside a string
`\\`	Backslash

14. Type System & Error Messages

Vect checks types **before** your program runs. If something is wrong, you get a clear message that points at the exact line and tells you in plain English what to fix.

Undefined variable

```
print(myVar)   # forgot to declare it
```

```
⚠  Type Error
Type error at line 1, col 7: 'myVar' is not defined.
Check the spelling or make sure you declared it with 'var'.
```

Type mismatch in an operation

```
print("price: " + 9.99)
```

```
⚠  Type Error
Type error at line 1, col 16: Cannot use '+' between a string and a 'float'.
To concatenate, convert to string first with str( ... ).
```

Wrong number of arguments

```
fn add(a: int, b: int) -> int { return a + b }
add(1, 2, 3)
```

```
⚠  Type Error
Type error at line 2, col 1: Function 'add' expects 2 argument(s) but got 3.
```

Type mismatch in declaration

```
var x: int = 3.14
```

```
⚠  Type Error
Type error at line 1, col 5: Variable 'x' is declared as 'int' but the value
is a float. Use 'int( ... )' to convert explicitly, or declare it as 'float'.
```

Type promotion — int and float mix automatically

You never need to manually convert when mixing `int` and `float` in arithmetic — Vect promotes the int to float automatically:

```
var result = 3 + 1.5    # fine — result is float 4.5
var area   = 2 * 3.14   # fine — area is float 6.28
```

---

15. CLI Reference

Command	What it does
`vect`	Start the interactive REPL
`vect run <file.vect>`	Compile and run a program
`vect check <file.vect>`	Type-check without running — find errors early
`vect ir <file.vect>`	Dump the generated LLVM IR — see inside the compiler

Examples

```
# Run the demo program
venv\Scripts\vect run examples\demo.vect

# Check your code for errors before running
venv\Scripts\vect check myprogram.vect

# See the LLVM IR the compiler generated
venv\Scripts\vect ir examples\fibonacci.vect

# Start exploring interactively
venv\Scripts\vect
```

---

16. Complete Example Programs

Example 1 — Fibonacci sequence

```
# fibonacci.vect
fn fib(n: int) → int {
    if n ≤ 1 { return n }
    return fib(n - 1) + fib(n - 2)
}

var i = 0
while i ≤ 10 {
    print(fib(i))
    i = i + 1
}
# Output: 0 1 1 2 3 5 8 13 21 34 55
```

Example 2 — Grade calculator

```
# grades.vect
fn letter_grade(score: float) -> string {
    if score ≥ 90.0 { return "A" }
    if score ≥ 80.0 { return "B" }
    if score ≥ 70.0 { return "C" }
    if score ≥ 60.0 { return "D" }
    return "F"
}

var scores = [95.0, 82.0, 67.0, 74.0, 55.0]
var total  = 0.0

for s in scores {
    total = total + s
}

var average = total / float(len(scores))
print("Average: " + str(average))
print("Grade: "   + letter_grade(average))
```

Example 3 — Linear algebra

```
# linear_algebra.vect
print("═══ Dot Product ═══")
var u = [3.0, 4.0, 0.0]
var v = [1.0, 0.0, 5.0]
print(u · v)    # 3.0

print("═══ Matrix Multiply ═══")
var A = [[2.0, 0.0], [0.0, 3.0]]   # scaling matrix
var B = [[1.0, 2.0], [3.0, 4.0]]
print(A @ B)    # [[2,4],[9,12]]

print("═══ Transpose ═══")
print(T(B))     # [[1,3],[2,4]]
```

Example 4 — Symbolic calculus

```
# calculus.vect
print("=== Differentiation ===")

sym f(x) = x**3 - 6*x**2 + 9*x
var df = d/dx(f(x))
print(df)                        # 3*x**2 - 12*x + 9

# Find where the slope is zero (critical points)
print(eval(df, x=1.0))          # 0.0  ← local max at x=1
print(eval(df, x=3.0))          # 0.0  ← local min at x=3

print("=== Physics ===")
sym s(t) = 30.0*t - 4.9*t**2    # throw upward at 30 m/s
var vel = d/dt(s(t))
print(vel)                        # 30.0 - 9.8*t
print(eval(vel, t=3.06))         # ≈ 0 — peak of flight
```

Example 5 — Everything together (the demo)

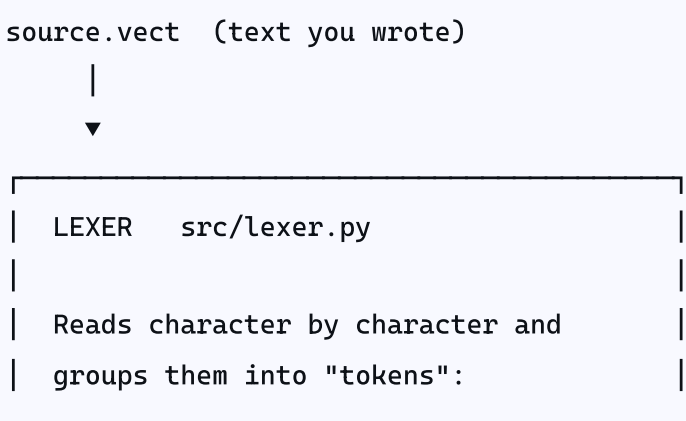
The file `examples/demo.vect` in the repository combines all features into one program designed to be run live. Run it with:

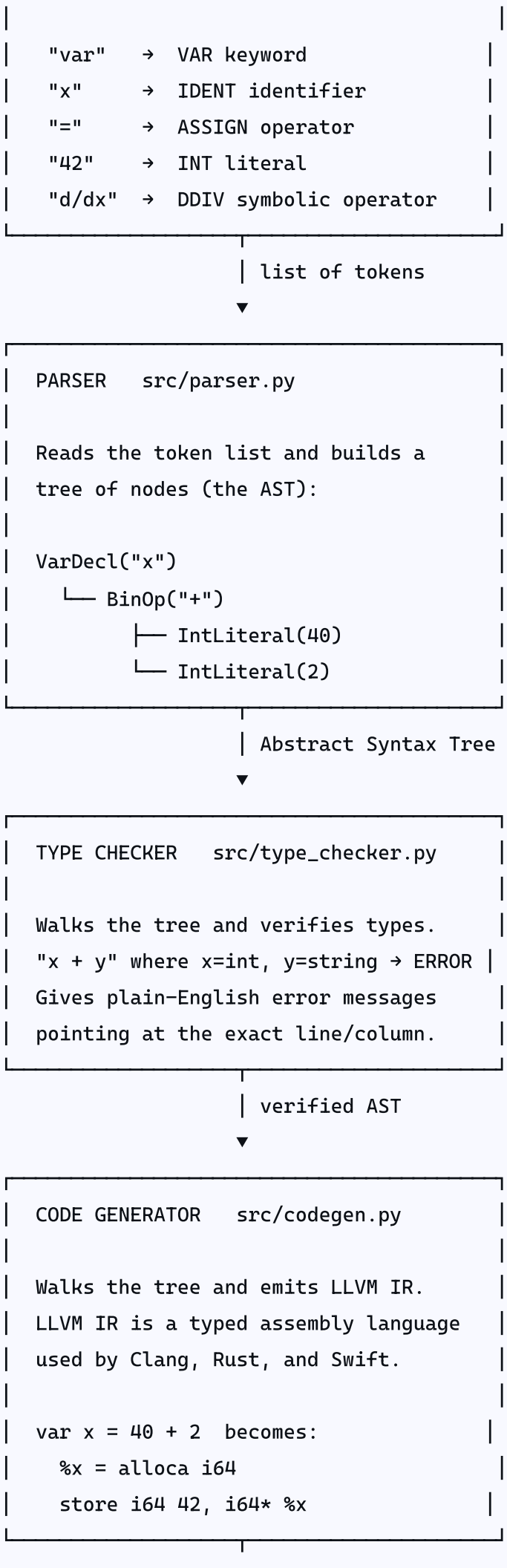
```
venv\Scripts\vect run examples\demo.vect
```

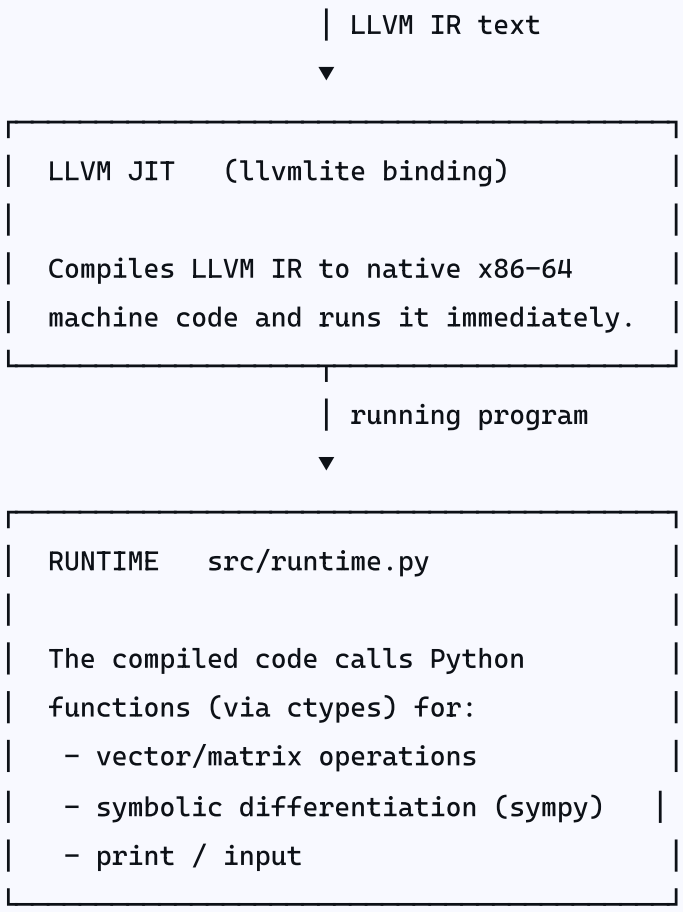
---

17. How the Compiler Works

You do not need to understand this to use Vect. But if you are curious — here is the full picture.




"var" → VAR keyword
"x" → IDENT identifier
"=" → ASSIGN operator
"42" → INT literal
"d/dx" → DDIV symbolic operator
list of tokens
PARSER src/parser.py
Reads the token list and builds a
tree of nodes (the AST):
VarDecl("x")
BinOp("+")
IntLiteral(40)
IntLiteral(2)
Abstract Syntax Tree
TYPE CHECKER src/type_checker.py
Walks the tree and verifies types.
"x + y" where x=int, y=string → ERROR
Gives plain-English error messages
pointing at the exact line/column.
verified AST
CODE GENERATOR src/codegen.py
Walks the tree and emits LLVM IR.
LLVM IR is a typed assembly language
used by Clang, Rust, and Swift.
var x = 40 + 2 becomes:
%x = alloca i64
store i64 42, i64* %x



The source files are all in the `src/` folder and are heavily commented. If you want to understand how any part works, open the file and read — it is written to be understood.

---

18. Quick Reference Card

Declarations

```
var name = value                    # variable
var name: type = value              # with explicit type
fn name(p: type) → type { ... }    # function
sym name(x) = expr                  # symbolic function
```

Types

```
int     42                          # whole number
float   3.14                        # decimal
bool    true  false                 # boolean
string  "hello"                     # text
vec     [1.0, 2.0, 3.0]             # vector
mat     [[1.0,2.0],[3.0,4.0]]       # matrix
```

Operators

```
+  -  *  /  %  **                   # arithmetic
==  ≠  <  ≤  >  ≥                   # comparison
and  or  not                        # logical
@                                   # matrix multiply
·                                   # dot product
d/dx                                # symbolic derivative
```

Control flow

```
if cond { } else if cond { } else { }
while cond { }
for x in vec { }
break   continue   return value
```

Symbolic calculus

```
sym f(x) = expr             # define symbolic function
var df = d/dx(f(x))         # differentiate
var n  = eval(df, x=2.0)    # evaluate numerically
```

CLI commands

```
vect                        # REPL
vect run  <file.vect>       # compile + run
vect check <file.vect>      # type-check only
vect ir   <file.vect>       # dump LLVM IR
```

Vect Language Guide — v0.1.0

Built by Eddie · github.com/Eddiegah/Vect

"Scientific computing should be syntax, not scaffolding."

---

19. v2 Features — What's New

VERSION 2.0

Five major features added: f-string interpolation, rich vec/mat stdlib, symbolic integration, native plot(), and AOT compilation to .exe.

19.1 — F-String Interpolation

Embed expressions directly inside strings using `f"..."` syntax. Any Vect expression can go inside the `{ }` braces.

```
var name  = "Eddie"
var score = 95
var pi    = 3.14159

print(f"Hello, {name}!")                # Hello, Eddie!
print(f"Score: {score}")                # Score: 95
print(f"Pi ≈ {pi}")                     # Pi ≈ 3.14159
print(f"Double score: {score * 2}")     # Double score: 190
print(f"{name} got {score}/100 — excellent!")
```

F-strings are expanded at compile time into a chain of string concatenations, so the codegen is unchanged — it's pure syntactic sugar.

19.2 — Rich Vector/Matrix Stdlib

```
# Vector operations
var v = [3.0, 4.0, 0.0]

print(norm(v))            # 5.0 — Euclidean magnitude
print(normalize(v))       # [0.6, 0.8, 0] — unit vector
print(zeros(4))            # [0, 0, 0, 0]
print(ones(3))             # [1, 1, 1]

var a = [1.0, 0.0, 0.0]
var b = [0.0, 1.0, 0.0]
print(cross(a, b))        # [0, 0, 1] — 3D cross product

# Matrix operations
var A = [[1.0, 2.0], [3.0, 4.0]]

print(det(A))             # -2.0 — determinant
print(inv(A))             # [[-2,1],[1.5,-0.5]] — inverse

# Solve the linear system Ax = b
var b2 = [5.0, 6.0]
var x  = solve(A, b2)
print(x)                  # solution vector
```

19.3 — Symbolic Integration

The companion to differentiation. `integral()` works the same way — pass a symbolic expression and get the antiderivative or the definite value.

```
# Definite integral — returns a float
sym f(x) = x**2
var area = integral(f(x), "x", 0.0, 3.0)
print(area)                 # 9.0  (area under x² from 0 to 3)

# Indefinite integral — returns symbolic antiderivative
var F = integral(f(x), "x")
print(eval(F, x=3.0))       # 9.0

# Physics: work done by variable force
sym force(x) = 2.0*x + 1.0
var work = integral(force(x), "x", 0.0, 5.0)
print(work)                 # 30.0 Joules

# Kinematics: position from velocity
sym velocity(t) = 20.0 - 9.8*t
var distance = integral(velocity(t), "t", 0.0, 2.041)
print(distance)             # ≈ 20.4 metres (to peak height)
```

19.4 — plot() Built-in

One line to visualise any symbolic function or data vector. The result is saved as `vect_plot.png` in your working directory.

```
# Plot a symbolic function
sym wave(x) = sin(x) * 2.0 + 0.5*sin(3.0*x)
plot(wave(x), x, -6.28, 6.28, "Wave function")

# Plot any curve — d/dx of a cubic
sym f(x) = x**3 - 3*x**2 + 2
plot(f(x), x, -1.0, 4.0, "Cubic polynomial")

# Plot data from vectors
var xs = [0.0, 1.0, 2.0, 3.0, 4.0]
var ys = [0.0, 1.0, 4.0, 9.0, 16.0]
plot_xy(xs, ys, "x squared data")
```

19.5 — AOT Compilation to .exe

Compile a Vect program to a standalone native executable. Requires gcc (MSYS2 on Windows, built-in on macOS/Linux).

```
# Compile fibonacci.vect to a native executable
vect build examples\fibonacci.vect -o fib

# Run it — no Python venv needed
.\fib.exe
0
1
1
2
3
5
8
13
21
34
55
```

How it works:

1. Parse + type-check the source
2. Generate LLVM IR (same as JIT)
3. Emit a native `.o` object file via llvmlite
4. Write a C shim that embeds Python for the runtime callbacks
5. Link with gcc → standalone `.exe`

---

20. Complete Quick Reference (v2)

Declarations

```
var name = value
var name: type = value
fn name(p: type) -> type { ... }
sym name(x) = expr
```

All operators

```
+  -  *  /  %  **          # arithmetic
==  ≠  <  ≤  >  ≥         # comparison
and  or  not               # logical
@                          # matrix multiply
·                          # dot product
d/dx                       # symbolic derivative (any variable)
```

All built-in functions

Function	Returns
`print(x)`	void
`input()`	string
`len(v)`	int
`sqrt / sin / cos / tan(x)`	float
`abs / floor / ceil(x)`	float / int
`int / float / str(x)`	converted
`range(n)`	vec
`T(m)`	mat (transpose)
`norm(v)`	float
`cross(a, b)`	vec
`normalize(v)`	vec
`zeros(n) / ones(n)`	vec
`det(A)`	float
`inv(A)`	mat
`solve(A, b)`	vec
`eval(expr, x=val)`	float
`integral(f, "x", lo, hi)`	float
`integral(f, "x")`	sym
`plot(f, x, lo, hi, "title")`	void (PNG)
`plot_xy(xs, ys, "title")`	void (PNG)

CLI commands

```
vect                            # REPL
vect run  <file.vect>           # JIT compile + run
vect build <file.vect> [-o exe] # AOT compile to .exe
vect check <file.vect>          # type-check only
vect ir   <file.vect>           # dump LLVM IR
```

Vect Language Guide — v2.0

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